

LC-3 Instruction Encodings

Arith./Logic	ADD [†]	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0						
		0	0	0	1	DR			SR1			0			0	0	SR2						
	ADD [†]	0	0	0	1	DR			SR1			1	imm5										
	AND [†]	0	1	0	1	DR			SR1			0			0	0	SR2						
	AND [†]	0	1	0	1	DR			SR1			1	imm5										
	NOT [†]	1	0	0	1	DR			SR			1	1	1	1	1	1						
Load/Store	LD [†]	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0						
		0	0	1	0	DR			PCoffset9														
	ST	0	0	1	1	SR			PCoffset9														
	LDI [†]	1	0	1	0	DR			PCoffset9														
	STI	1	0	1	1	SR			PCoffset9														
	LDR [†]	0	1	1	0	DR			BaseR			offset6											
	STR	0	1	1	1	SR			BaseR			offset6											
Branch	JMP	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0						
		1	1	0	0	0			0	0	BaseR		0						0	0	0	0	0
	BR	0	0	0	0	cn	cz	cp	PCoffset9														
	JSRR	0	1	0	0	0			0	0	BaseR		0						0	0	0	0	0
	JSR	0	1	0	0	1	PCoffset11																
Misc.	LEA	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0						
		1	1	1	0	DR			PCoffset9														
	TRAP	1	1	1	1	0			0	0	0	trapvect8											
	RTI	1	0	0	0	0												0	0	0	0	0	0
	<i>illegal</i>	1	1	0	1	x			x	x	x	x	x	x	x	x	x						

[†] Updates condition code (CC)

LC-3 Memory Map

